

Education

Habib University

BS in Computer Science

August 2021 – May 2025

- CGPA: 4.0/4.0
- **Minors:** Electrical & Computer Engineering, Mathematics
- President's Medal, Dean's Medal (Computer Science)
- 100% 4-year Merit Scholarship
- Dean's List (Spring'22, Fall'22, Spring'23, Fall'23, Spring'24, Fall'24)
- President's List (2022, 2023, 2024)
- High Academic Achievement Scholarship (Fall'22, Spring'23, Fall'23, Fall'24)

Stanford University

International Honors Program

June 2023 – August 2023

- GPA: 3.887/4.3
- **Coursework:** Machine Learning (CS229), High Performance Computing, Technological Entrepreneurship

Experience

Autonomous Vehicle Algorithms Engineer

Covolv.ai

Karachi, Pakistan

June 2025 - Present

- Responsible for development, optimization and deployment of ML-based algorithms for autonomous driving products for OEMs.

Teaching Assistant

Habib University

Karachi, Pakistan

August 2022 - April 2025

- Teaching Assistant for the course Programming Fundamentals (Fall 2022), Discrete Mathematics (Spring 2023), Computer Architecture (Spring 2024), Linear Algebra (Spring 2024), Deep Learning (Fall 2024), Algorithm Design & Analysis (Spring 2025)
- Responsibilities include assisting students in lab, conducting office hours, conducting recitations, designing and grading course assignments.
- Best Teaching Assistant award for Spring 2024 by the Office of Career Services, Habib University.

Undergraduate Research Assistant

Habib University

Karachi, Pakistan

January 2024 - August 2024

- Developed a deep learning based pipeline for identification of CRISPR Cas 13b based guide potential crRNAs for treating viral infections.
- Presented poster titled "Designing towards an informed CRISPR Cas13b based drug design"

Research Intern (ML)

Texas A&M University

College Station, Texas

May 2024 - August 2024

- Implemented current spectral calibration methods for Swept Source OCT (SS-OCT) systems based on literature and industry practices.
- Developed a deep learning-based spectral calibration pipeline using LSTMs, reducing calibration error to 12%.
- Enhanced system robustness to jitter noise and signal down-sampling, improving overall accuracy and reliability
- Presented poster titled "Enhancing Spectral Calibration of SS-OCT: Using LSTM Architecture for Improved Noise Handling"

Open Source Contributions

Autoware - Autonomous Highway Pilot

Autoware Foundation

- Collaborating to build an open-source highway autonomy system that can power safe, level-4, autonomous driving around the world.

Conference Presentations

Fine-tuned LLM based approach to Scientific Text Simplification

September 2024

Presented at CLEF 2024 Lab Proceedings held in Grenoble, France

S.M Ali H. Sajid, O. Aijaz, O. Waheed, F. Alvi, A. Samad

<https://ceur-ws.org/Vol-3740/paper-309.pdf> [🔗](#)

Projects

KisanEye - Aerial-based Crop Health Monitoring System

- Built a Pixhawk-RPI-based quadrotor for crop health monitoring through Vision and NDVI based detection using RGB camera and NIR filters for local farms in Sindh, Pakistan.
- Developed a mobile application interface for real-time monitoring, command, and control of the quadrotor.
- This project was completed as part of the Farm Robotics Challenge 2025.
- Technologies: Python, C++, React-native, PX4, Pixhawk, Raspberry Pi

Phantom X Pincher PickPlace Package

- Designed and implemented a custom perception-based pick-and-place pipeline for the Phantom X 4DOF robotic arm.
- Developed the entire pipeline, without relying on external kinematics, motion planning, or control libraries/frameworks.
- Technologies: Matlab, Python.

ALIF: Series of Pretrained Urdu Generative Models

- Curated a novel dataset of 6 billion tokens for pretraining urdu language models.
- Developed BPE based Urdu monolingual and Urdu-English bilingual tokenizers, for efficient tokenization.
- Developed instruct tuning datasets for urdu language model.
- Developed, trained, and instruction-tuned decoder-based language models (100M and 1B parameters) for the Urdu language, outperforming multilingual models of comparable sizes.
- Technologies: Python, Pytorch, accelerate, Megatron

ENVISION for Visually Impaired Individuals

- Developed an accessible mobile digital banking application for Faysal Bank, specifically designed for visually impaired users.
- Secured First Prize at the Faysal Bank FinTech Innovation Hackathon 2024, winning a cash prize of PKR 500,000.
- Technologies: React Native, Node.js.

Indoor Navigation Assistance Smart Walking Cane

- Designed a smart walking cane mobile robot to assist individuals with visual impairments in navigation.
- Implemented PID control to ensure smooth operation and improved user experience.
- Technologies: ROS, Gazebo, MATLAB.

Extra-Curricular

Set Alumni Network (SAN) — Core Team Member

The SET School

January 2023 - Present

- Responsible for managing alumni network's website and social media.
- Managed logistics and hosted the annual SET alumni reunion event 2023.
- Conducted session at The SET School titled "Modern Trends in Machine Learning".

Vice President — Maths Club

Habib University

May 2023 - May 2024

- Led a team of 8+ members to conduct workshops, talks and Math Circles.
- Co-led the core team to conduct Mathema'2023, Habib University's flagship Maths Olympiad for High School students.

Technology Instructor — Serve Club

Habib University

August 2022 - Dec 2022

Technologies

Languages: C++, C, Python, Matlab, Javascript, SQL, Verilog HDL

Softwares/Frameworks: Simulink, CUDA, Creo, PyTorch, OpenCV, Node.js, accelerate, Megatron, PyCuda, ROS2

Microcontrollers: Arduino, Raspberry PI, Nvidia Jetson, Nvidia AGX, Qualcomm SoCs